

Case Study Assignment (CMP370: Internet of Things)

Title

Case Study Assignment: Analysis and Design of an Internet of Things System

Course: CMP370 – Internet of Things

Program: BE Information Technology

Total Marks: 20

Word Limit: 2500–3000 words

Objective

After completing the laboratory exercises and research work, students are expected to critically analyze one real-world IoT application and relate it to the theoretical concepts and practical implementations covered in this course.

The case study should not merely describe an IoT project; instead, it should explain how the project is designed, why particular technologies were selected, what challenges exist, and how the system can be improved.

Step 1: Select One Topic

Choose **one** project from your previously submitted research work (10 IoT Projects).

Examples include

- Smart Weather Monitoring System
- Smart Irrigation System
- Smart Home Automation
- Smart Parking System
- Smart Healthcare Monitoring
- Smart Waste Management
- Smart Street Lighting
- Smart Energy Meter
- Industrial IoT Monitoring
- Air Quality Monitoring System

Case Study Structure

Cover Page

- Course Name
- Course Code
- Case Study Title
- Student Name
- Roll Number
- Instructor Name (Er. Rishi K. Marseni)
- Department
- Semester
- Date

Table of Contents

1. Introduction (250–300 words)

Introduce

- Internet of Things
- Importance of the selected application
- Current trends
- Motivation for choosing the topic

Explain why the selected project is important in today's world.

2. Background Study (300–400 words)

Explain

- Existing problem
- Current solutions
- Need for IoT

Support the discussion using at least **3 research papers or credible online sources**.

3. Problem Statement (150–200 words)

Clearly define

- What problem does this IoT system solve?
- Why is the problem important?
- Who benefits from the system?

4. System Architecture (400–500 words)

Explain the architecture using the concepts learned in Unit 2.

Include

Device Layer

Examples

- ESP32
- Raspberry Pi
- Arduino
- DHT11
- Soil Moisture Sensor
- Relay Module

Network Layer

Explain

- WiFi
- MQTT
- HTTP
- REST API
- WebSocket

Cloud Layer

Explain

- AWS EC2
- Google Cloud
- ThingSpeak
- Firebase

Students should include an architecture diagram.

5. Hardware Components (250–300 words)

Discuss every hardware component.

Example (SOC/SOB)

ESP32(or any other MCU/MPU)

- Features
- Advantages
- Memory
- GPIO
- WiFi

Sensors

- Working Principle

- Accuracy
- Applications

Actuators

- Relay
- Pump
- Fan
- Motor

6. Software Components (300–350 words)

Explain

Programming Languages

- Python
- C++
- Arduino IDE

Frameworks

- FastAPI
- Flask

Database

- TinyDB
- SQLite
- MySQL
- PostgreSQL

Dashboard

- HTML
- JavaScript
- Chart.js

Cloud

- AWS EC2
- Microsoft Azure
- Google Cloud Computing

7. Communication Protocols (250–300 words)

Discuss

- ☐ HTTP
- ☐ MQTT
- ☐ CoAP

- ☐ REST API
- ☐ WebSocket

Compare

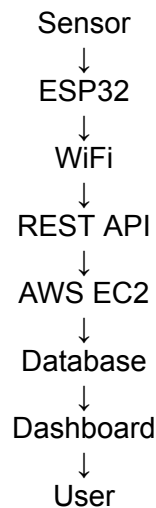
- Speed
- Reliability
- Power Consumption
- Scalability

Explain why the selected protocol is suitable.

8. Data Flow Analysis (250–350 words)

Explain the complete flow.

Example



Relate this section directly to Lab 2, Lab 3, and Lab 5.

9. Security Analysis (250–300 words)

Discuss

Possible threats

- Unauthorized access
- Data tampering
- Replay attack
- Password attack

Security Measures

- HTTPS
- SSL/TLS
- Authentication
- API Keys

- Encryption

Relate to Unit 6.

10. Mapping with Course Labs (300–350 words)

This section is compulsory.

Explain how your selected project relates to each laboratory.

Example

Lab 1

- Hosting web server using AWS EC2
- Connection with project
- Web dashboard deployment.

Lab 2

- REST API
- Connection
- Sensor uploads data through FastAPI.

Lab 3

- Dashboard
- Connection
- Real-time visualization.

Lab 4

- Embedded Devices
- Connection
- ESP32 and sensors.

Lab 5

- End-to-End IoT
- Connection
- Complete workflow from sensor to cloud.

11. Challenges and Limitations (200–250 words)

Possible issues

- Internet dependency
- Sensor accuracy
- Security
- Power consumption

- Cloud cost
- Scalability
- Maintenance

12. Future Improvements (200–250 words)

- ☐ Discuss
- ☐ AI integration
- ☐ Machine Learning
- ☐ Edge Computing
- ☐ 5G
- ☐ Digital Twins
- ☐ Predictive Analytics
- ☐ Blockchain

13. Personal Reflection (150–200 words)

Discuss

- ☐ What you learned
- ☐ Skills gained
- ☐ How labs helped
- ☐ Future interest

14. Conclusion (150–200 words)

Summarize

- Key findings
- Importance of IoT
- Future impact

15. References

Minimum (10 references)

Use IEEE format.

- ☐ Research papers
 - ☐ Books
 - ☐ Official documentation
 - ☐ Reliable websites
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Figures Required

Students should include at least

- IoT Architecture Diagram
- Data Flow Diagram
- System Block Diagram
- Dashboard Screenshot
- Sensor Connection Diagram
- Actuator Connection Diagram